WORKPLACE INJURIES

IN ITALY'S TRANSPORTATION & STORAGE SECTOR

Final Report

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Amazon Data Analytics Cohort 18

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Document Control

Version	Date	Author	Description
0.1	Feb. 13, 2026	Giulio Galbiati	Initial document structure created based on the C1 template; defined main sections, added project description and scope.
0.2	Feb. 24, 2026	Giulio Galbiati	Formatting aligned with the Data Curation document; added sections on data sources, data preparation, and database design.
0.3	Mar. 11, 2026	Giulio Galbiati	Integrated exploratory data analysis results and analytical findings; added sections on key findings, conclusions, and limitations.
0.4	Mar. 21, 2026	Giulio Galbiati	Added section with description of Datafolio design.
0.5	Mar. 29, 2026	Giulio Galbiati	Added other communication components dashboard description, and AI tools disclosure included.
0.6	Mar. 30, 2026	Giulio Galbiati	Added charts and final review
1.0	Mar. 30, 2026	Giulio Galbiati	First official release submitted for assessment; content frozen for evaluation.

Document Status

This document represents the final report prepared for the **Amazon Data Analytics Program**.

It summarizes the data sources used, the data preparation and database design process, and the exploratory analysis conducted on the curated dataset. The report also presents the **main analytical findings** and discusses their implications, together with concluding **remarks and potential directions for future work**.

If any updates are needed in the future, they will be issued as a new version and recorded in the Document Control table to keep track of changes.

Project Description

Project: Work-related injuries in the Italian transportation and storage sector

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Executive Summary

This report analyzes work-related injuries in the Italian Transportation and Storage sector (NACE Rev. 2 – Activity H) using EUROSTAT data for 2014–2023. The objective is to identify **structural risk patterns** and **support targeted, evidence-based safety decisions**.

Injury risk in the sector is structurally higher than the national average and has remained broadly stable over time. When adjusted for workforce size, Italy is aligned with comparable European economies, indicating that risk is driven by the nature of logistics operations rather than country-specific factors.

Risk is not widespread — it is concentrated. A **limited number of recurring operational contexts account for most injuries**, particularly:

- movement-related activities
- transport operations
- goods handling

These patterns are consistent across countries and over time, confirming that **occupational risk is predictable and driven by repeatable mechanisms**.

This is not primarily a compliance issue. It is an operational issue.

Implications for management are clear:

- focus on high-frequency risk scenarios
- redesign processes where risk originates
- reinforce execution through training and standard routines
- avoid broad, non-targeted safety initiatives

The analytical framework developed in this report provides a **clear and consistent basis to identify priority areas and support data-driven safety decisions** in transportation and logistics operations.

1. Introduction

1.1 Problem Statement

Work-related injuries remain a significant occupational safety concern in the transportation and storage sector (**NACE Rev. 2 – Activity H**). International and European safety authorities consistently classify this industry among those with **the highest exposure to workplace accidents** due to the operational characteristics of logistics activities, such as goods handling, vehicle operations, and time-sensitive workflows.

Although regulatory frameworks and workplace safety policies have been progressively strengthened across Europe, accident events continue to occur. **Compliance alone may not be sufficient to fully understand and mitigate operational safety risks**.

In many organizations, injury records are systematically collected to satisfy regulatory reporting requirements. However, these datasets are **not always fully leveraged to identify structural patterns, recurrent risk conditions, or operational factors associated with accidents**. As a result, valuable information that could support **preventive safety management** may remain underutilized.

From an operational management perspective, understanding **how injuries occur - including which job roles are most affected, under what circumstances accidents happen, and which demographic groups are more exposed** - can provide important insights for improving workplace safety and prioritizing preventive actions.

Given the structural importance of the Italian transportation and storage sector within the **European logistics network**, analyzing historical accident statistics offers an opportunity to better understand the dynamics of occupational injuries in this industry.

This project therefore analyzes **work-related injury data for the Italian transportation and storage sector using EUROSTAT statistics covering the period 2014–2023**. The analysis focuses on identifying patterns in **injury incidence, accident characteristics, and workforce exposure using comparative and normalized indicators**.

1.2 Potential Impact of the Analysis

The objective of this analysis is not only descriptive but also interpretative: by examining historical accident data through a **structured analytical approach**, the study aims to support a more **evidence-based understanding of occupational safety conditions within the sector**.

Specifically, the analysis seeks to:

- Identify **structural patterns in injury occurrence over time**.
- Highlight **operational contexts and job roles associated with higher accident exposure**.
- Examine **demographic characteristics of injured workers**, such as age groups and gender distribution.

- Benchmark the **Italian transportation and storage sector against comparable European economies**, providing a broader contextual perspective.

From a managerial standpoint, these insights can contribute to a **more informed prioritization of safety initiatives**, helping organizations focus resources on areas where **risk exposure appears to be structurally higher**.

While this study does not aim to establish causal relationships, the patterns identified through the analysis can support **data-informed discussions on workplace safety, risk prevention strategies, and policy evaluation within the logistics sector**.

1.3 How to Read This Report

This report is structured to progressively move from **data description to analytical interpretation and practical implications**.

- Sections 2 and 3 describe the **data sources and preparation process**, providing transparency on the construction of the analytical dataset.
- Section 4 presents the **exploratory data analysis**, highlighting key patterns in occupational injuries across countries, sectors, and operational contexts.
- Section 5 interprets these patterns and discusses their **implications for understanding occupational risk in the transportation and storage sector**.
- Sections 6 and 7 introduce the **datafolio and interactive dashboard**, which allow the results of the analysis to be explored visually.
- Sections 8 and 9 summarize the **main conclusions, potential future research directions, and limitations of the study**.
- Finally, Section 10 provides **references**, **methodological note**s, and a **disclosure on the use of Artificial Intelligence (AI) tools**.

2. Data Sources

The analysis is based on publicly available statistical data provided by Eurostat, the statistical office of the European Union. The datasets originate from the European Statistics on Accidents at Work (ESAW) domain and from the Labour Force Survey (LFS) domain.

ESAW datasets provide detailed information on occupational accidents, including demographic characteristics, severity, and accident circumstances. Employment statistics from the Labour Force Survey are used as the exposure denominator to compute normalized indicators expressed as casualties per 1,000 employees.

The primary data source used in this study is:

- **Eurostat - Accidents at work: statistics on causes and circumstances**

  https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Accidents_at_work_-_statistics_on_causes_and_circumstances#Data_sources

2.1 ESAW Accident Circumstance Datasets

The following Eurostat datasets provide detailed information on accident circumstances within the Transportation and Storage sector (NACE Rev. 2 – Activity H):

- **hsw_ph3_01**

  Accidents at work by sex, age, severity, activity and workstation

  https://ec.europa.eu/eurostat/databrowser/view/hsw_ph3_01__custom_18403289/bookmark/table?lang=en&bookmarkId=7db12bdf-a43f-41b6-9a8d-78dbddfe2ea0&c=1760438842000

- **hsw_ph3_02**

  Accidents at work by sex, age, severity, activity and working environment

  https://ec.europa.eu/eurostat/databrowser/view/hsw_ph3_02__custom_18403291/bookmark/table?lang=en&bookmarkId=2bd7cfeb-2189-4c2c-b72d-63b213b81335&c=1760438854000

- **hsw_ph3_03**

  Accidents at work by sex, age, severity, activity and working process

  https://ec.europa.eu/eurostat/databrowser/view/hsw_ph3_03__custom_18403293/bookmark/table?lang=en&bookmarkId=edb75a76-a88a-452c-b453-e87d5eccaf9b&c=1760438870000

- **hsw_ph3_04**

  Accidents at work by sex, age, severity, activity and specific physical activity

  https://ec.europa.eu/eurostat/databrowser/view/HSW_PH3_04__custom_3560699/bookmark/table?lang=en&bookmarkId=4163e75b-1dec-4f45-8fcd-36eab71076e9&c=1665503705938

- **hsw_ph3_06**

  Accidents at work by sex, age, severity, activity and deviation

  https://ec.europa.eu/eurostat/databrowser/view/HSW_PH3_06__custom_3227390/bookmark/table?lang=en&bookmarkId=dc99b46e-711f-46e6-bc90-99cf81f00426&c=1660924613321

- **hsw_ph3_08**

  Accidents at work by sex, age, severity, activity and contact mode of injury

  https://ec.europa.eu/eurostat/databrowser/view/HSW_PH3_08__custom_3227437/bookmark/table?lang=en&bookmarkId=e5d79b4f-566e-4dee-84a2-a98f440e5a57&c=1665504724298

These datasets provide detailed accident descriptions covering variables such as workstation, working environment, working process, specific physical activity, deviation, and contact mode of injury.

2.2 Aggregated Accident Datasets

Two additional ESAW datasets provide aggregated accident totals:

- **hsw_n2_01**

  Non-fatal accidents at work by NACE Rev. 2 activity and sex

  https://ec.europa.eu/eurostat/databrowser/view/HSW_N2_01__custom_19348532/default/table

- **hsw_n2_02**

  Fatal accidents at work by NACE Rev. 2 activity

  https://ec.europa.eu/eurostat/databrowser/view/hsw_n2_02__custom_19348877/default/table

These datasets were used to compute total accident counts and severity distributions.

2.3 Employment Dataset

Employment data were obtained from the Eurostat Labour Force Survey and were used as the denominator for computing normalized injury indicators.

- **lfsa_eisn2**

  Employed persons by occupation and economic activity (NACE Rev. 2)

  https://ec.europa.eu/eurostat/databrowser/view/lfsa_eisn2__custom_19447243/default/table

This dataset provides the number of employed persons by economic activity and year, allowing injury counts to be expressed relative to workforce size.

2.4 Data License

All datasets used in this study are publicly available under the Creative Commons **Attribution 4.0 International (CC BY 4.0) license**.

License information: https://creativecommons.org/licenses/by/4.0/

This license permits sharing and adaptation of the data, provided that appropriate credit is given to the original source.

3. Data Preparation and Database Design

The datasets used in this project were obtained from multiple Eurostat statistical sources belonging to the European Statistics on Accidents at Work (ESAW) and the Labour Force Survey (LFS) domains.

Because these datasets originate from different statistical systems and extraction formats, a structured data preparation process was required before they could be analyzed jointly.

The preparation workflow follows a simplified extract–transform–load (ETL) logic in which datasets are first examined and cleaned, then harmonized and integrated into a relational analytical database.

The data preparation workflow consisted of **two main phases**:

- **Dataset profiling, cleaning, and structural harmonization** within a spreadsheet environment
- **Integration of the curated datasets into a relational analytical database** for multidimensional analysis

The **first phase** focused on understanding, cleaning, and harmonizing the datasets, while the **second phase** involved building a relational database designed to support consistent and reproducible analysis of occupational injury patterns.

3.1 Data profiling

Data profiling was conducted in a spreadsheet environment to examine the structure and analytical content of the downloaded datasets before they were integrated into the analytical workflow.

Profiling focused on understanding dataset dimensions, variable definitions, and the role of each attribute within the Eurostat statistical structure. Attention was given to distinguishing analytical variables (such as country, activity, severity, sex, age, and accident circumstance categories) from auxiliary metadata fields included in the original exports.

This exploratory inspection allowed the identification of duplicated descriptive fields, administrative attributes without analytical relevance, and potential structural inconsistencies between datasets originating from different Eurostat extraction views.

3.2 Data cleaning

Data cleaning was performed to simplify and standardize the datasets. These operations were conducted primarily within the spreadsheet environment prior to database integration.

Non-analytical metadata fields - including confidentiality flags, status indicators, and redundant descriptive labels - were removed to reduce dataset complexity and improve readability. Column names were standardized to ensure consistent naming conventions across datasets and to facilitate subsequent processing steps.

During this phase, attention was also given to identifying structural zero values associated with reporting gaps or suppressed observations. Where appropriate, these values were treated as missing data rather than valid measurements to avoid distortions in subsequent analysis.

3.3 Structural harmonization

Structural harmonization was required because accident statistics and employment statistics originate from different Eurostat statistical domains.

This process included the standardization of country selection based on employment weight and the alignment of **NACE Rev. 2** classifications across injury and employment datasets. It also involved consolidating six accident "circumstance" classifications into a unified categorical structure.

These steps ensured that datasets could be combined using **shared analytical dimensions** such as **country, year, activity, severity, sex, and accident circumstance**, allowing consistent aggregation and comparison across datasets.

3.4 Derived indicators

Derived normalized indicators were constructed using employment data from the Labour Force Survey in order to enable meaningful comparison across countries with different workforce sizes. The principal metric used in the analysis is **casualties per 1,000 employees**, calculated by dividing the number of recorded accidents by the number of employees in the corresponding sector and year.

This normalization transforms raw accident counts into incidence indicators that allow comparison of occupational risk across countries and economic activities.

3.5 Analytical Database Construction

Following the spreadsheet-based preparation phase, the curated datasets were imported into a **relational SQLite database** designed to support consistent and reproducible analysis of work-related injuries.

The database was structured to integrate accident statistics and employment data within a unified analytical framework. Its primary objective is to enable multidimensional analysis across countries, time periods, demographic groups, and accident circumstances while preserving traceability to the original statistical sources.

The database adopts a **multi-star schema architecture**, selected because the project integrates several datasets with different levels of analytical granularity. The structure consists of five fact tables containing quantitative observations and seven dimension tables providing standardized classifications such as country, economic activity, sex, age, severity, and accident circumstance categories.

Each fact table retains its natural statistical resolution to avoid artificial aggregation. The most detailed table describing accident circumstances follows the analytical grain:

Country × Year × Activity × Circumstance × Severity × Sex × Age.

Other fact tables store aggregated accident totals and employment data used as the exposure denominator for normalized indicators.

This relational design enables flexible aggregation and comparison across datasets while ensuring consistency between injury counts and employment statistics. A comprehensive technical description of

the database schema, fact tables, dimension tables, and data lineage is provided in the Data Curation report submitted during the previous phase of the project.

4. Exploratory Data Analysis: Patterns and Comparisons

This section presents the **exploratory data analysis (EDA)** of work-related injuries in the **Transportation and Storage sector (NACE Rev. 2 – Activity H)**. The analysis adopts a comparative framework based on normalized indicators expressed as **casualties per 1,000 employees**. The objective is to identify structural patterns in occupational risk, assess Italy's position within the European context, and benchmark the sector against the broader Italian economy.

Key Takeaways

- **Italy's injury incidence is broadly aligned with comparable European economies.**
- **The Transportation and Storage sector shows higher injury incidence than the Italian national average.**
- **Workplace injuries concentrate in recurring operational contexts such as movement, transport operations, and goods handling.**
- **Absolute accident counts are strongly influenced by workforce size,** making normalized indicators essential for comparison.

4.1 Employment Exposure and Workforce Structure

How is employment distributed across the European transportation and storage sector?

Employment in the Transportation and Storage sector is concentrated in a limited number of large European economies and remains structurally male-dominated.

Before analyzing injury patterns, it is important to understand the workforce exposed to occupational risk. Employment in NACE H is concentrated in a limited number of large European economies and has remained relatively stable over time. The sector workforce is also strongly male-dominated, a characteristic that provides important context for interpreting injury patterns and gender differences in incidence rates.

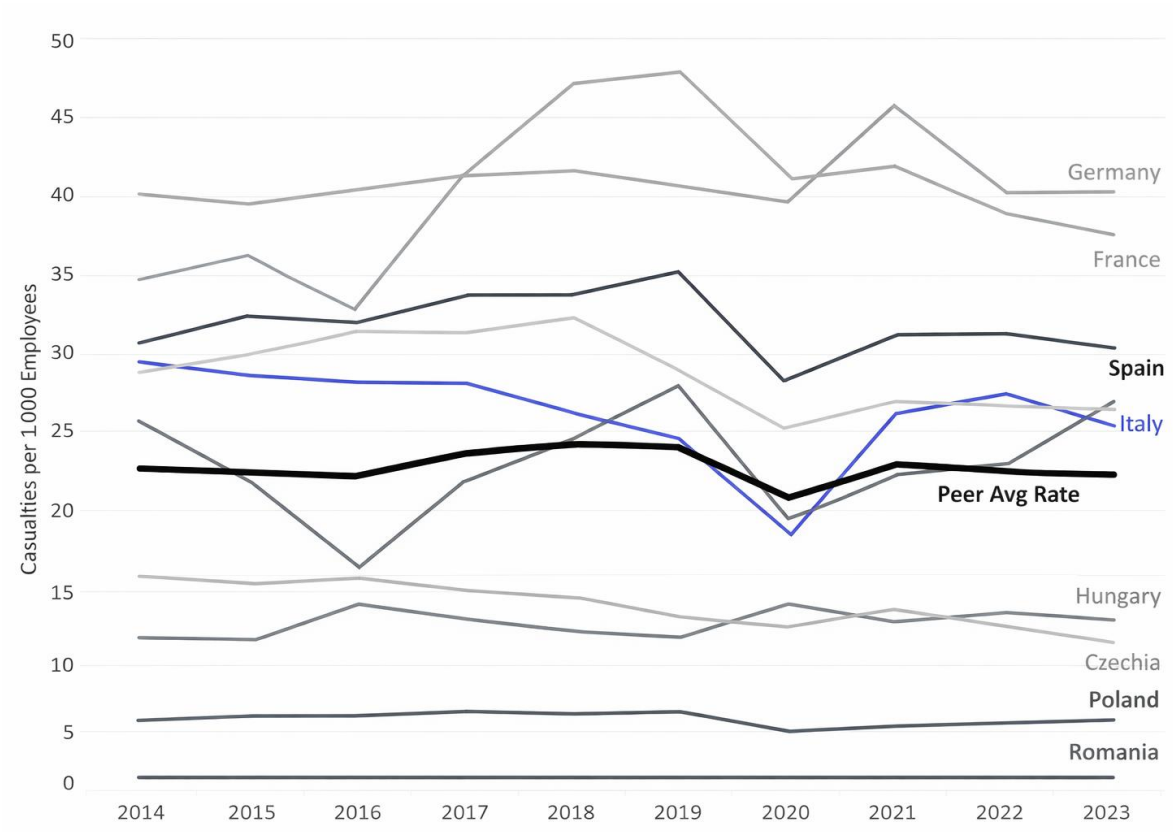
4.2 Cross-Country Comparison of Injury Incidence

How does Italy compare with other European countries?

Italy's injury incidence in the Transportation and Storage sector is broadly aligned with comparable European economies.

Absolute accident counts vary substantially across countries, primarily reflecting differences in workforce size. When casualty figures are normalized by employment exposure, cross-country differences become more limited and more meaningful from a risk perspective. Italy does not emerge as an outlier in relative injury incidence and remains broadly aligned with the European peer group. Injury incidence shows moderate year-to-year variation but no clear long-term trend over the 2014–2023 period.

Figure 4.1 - Injury Incidence Trend in Transportation and Storage (EU Top-10, 2014–2023)



Injury incidence, measured as casualties per 1,000 employees, remains broadly stable over time across major European transportation and storage economies.

Year-to-year fluctuations are present, but no persistent upward or downward trend emerges, indicating that occupational risk in the sector is structurally stable rather than cyclical.

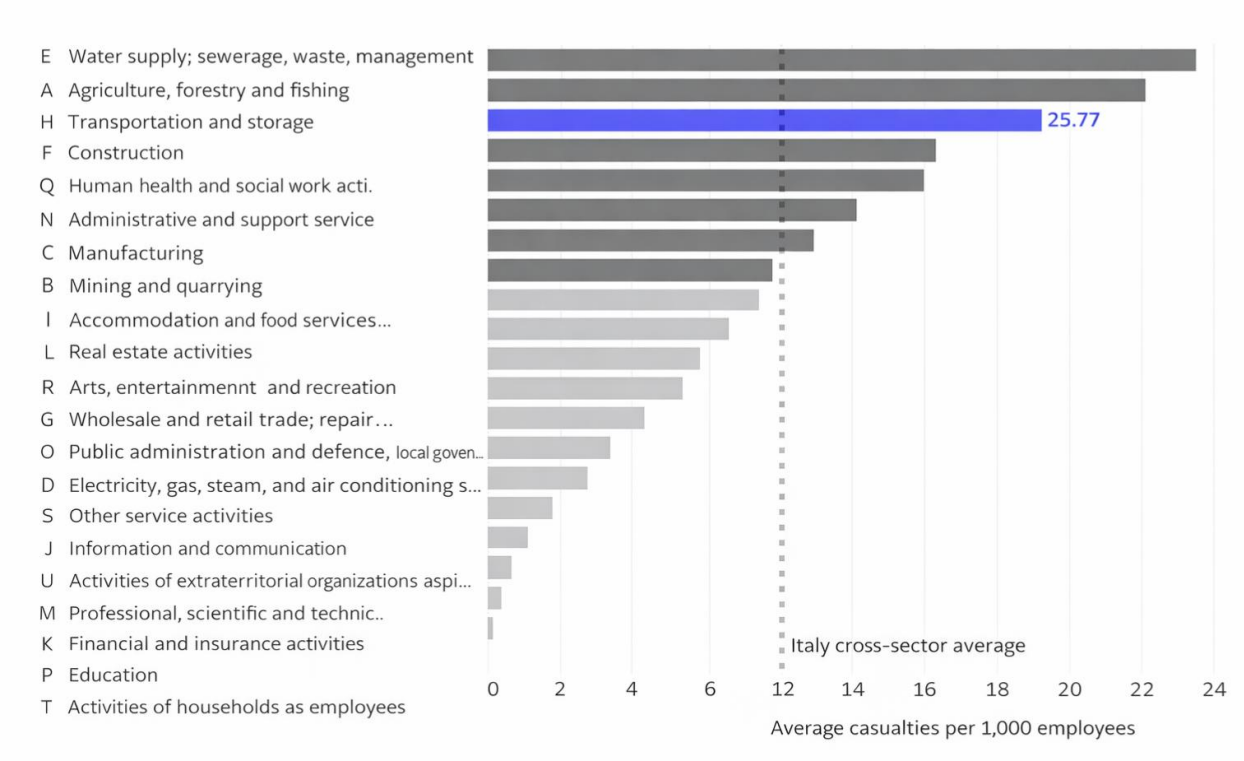
4.3 Sector Benchmark within the Italian Economy

How does the Transportation and Storage sector compare with other economic sectors in Italy?

Within Italy, the Transportation and Storage sector records higher injury incidence than the national average but does not represent the most extreme risk environment among economic activities.

To better understand the sector's relative risk profile, injury incidence was benchmarked against the overall Italian economy. The results indicate that the Transportation and Storage sector consistently records higher injury incidence than the national average across economic activities. However, the sector remains within the broader distribution of occupational risk across industries and should not be interpreted as an outlier.

Figure 4.2 - Injury Incidence by Economic Activity in Italy (Average 2014–2023)



Injury incidence varies significantly across economic activities when normalized by employment.

The Transportation and Storage sector consistently records higher-than-average injury incidence within the Italian economy, confirming its classification as a relatively high-risk activity, though not the most extreme.

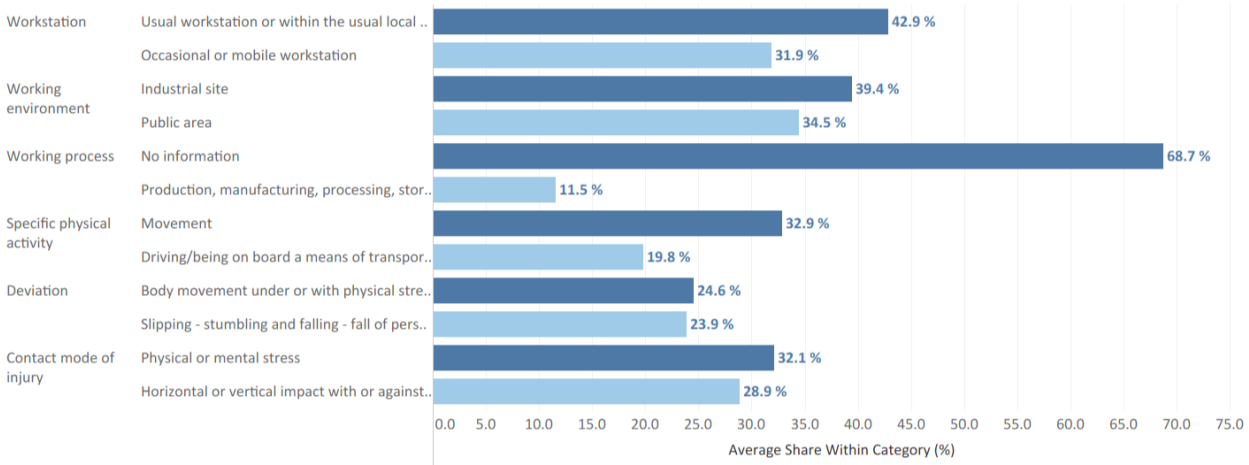
4.4 Operational Circumstances of Workplace Injuries

In which operational contexts do most accidents occur?

Workplace injuries in the sector tend to concentrate in a limited set of recurring operational contexts.

The circumstance-level analysis shows that accidents are most frequently associated with worker movement, transport operations, goods handling activities, and impact-related events. These patterns appear consistently across countries, and Italy's circumstance profile broadly mirrors that observed in the European peer group.

Figure 4.3 - Distribution of Workplace Injuries by Operational Circumstance (NACE H, EU Top-10, 2014–2023)



Workplace injuries are concentrated in a limited number of recurring operational contexts, particularly movement, transport operations, and goods handling.

This concentration highlights that occupational risk in the sector is driven by repeatable and identifiable mechanisms rather than diffuse or random factors.

4.5 Relationship Between Workforce Size and Accident Counts

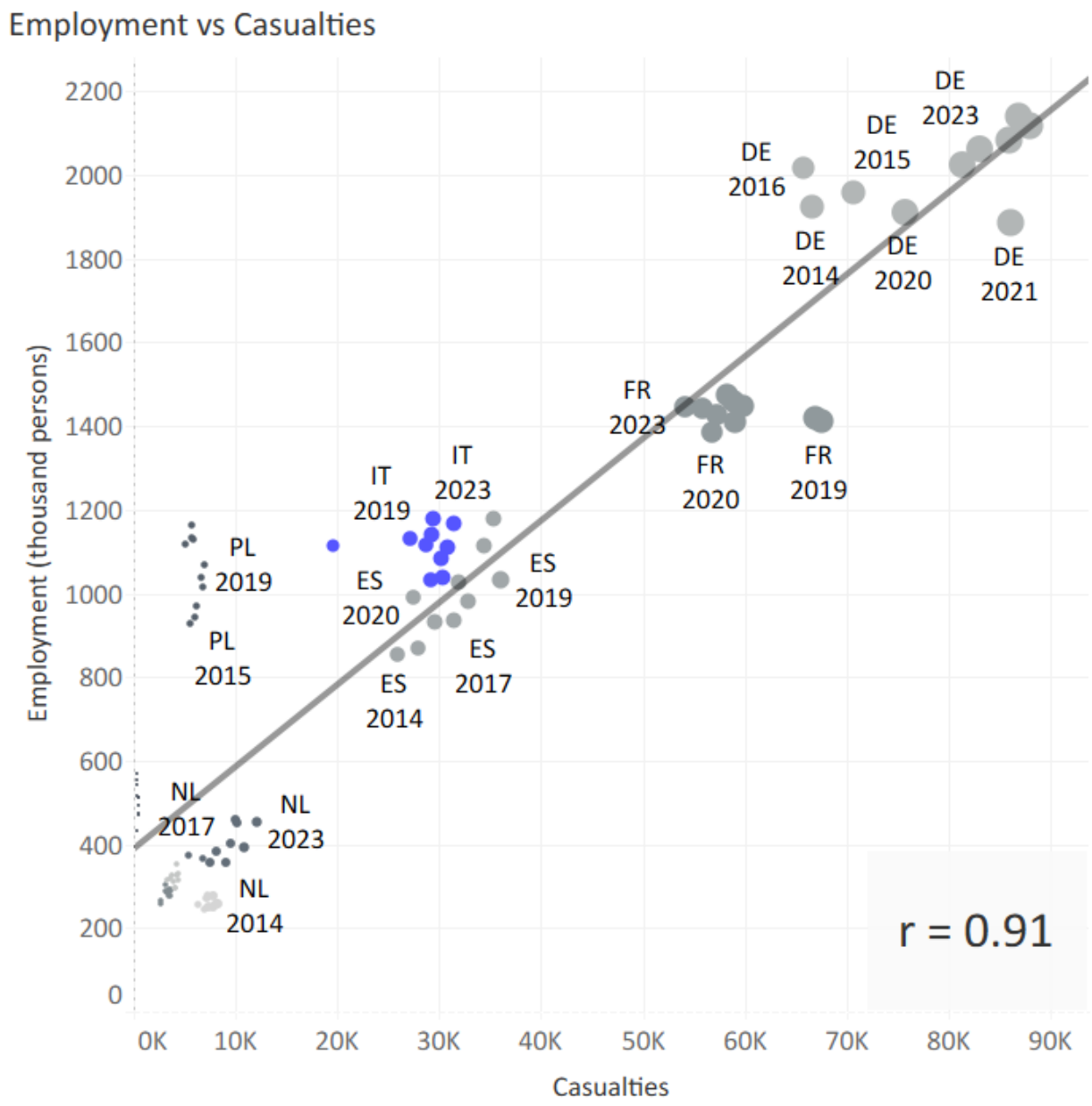
How does workforce size influence accident counts?

Accident counts are strongly influenced by workforce size, making normalized indicators essential for meaningful risk comparison.

The analysis examined the relationship between workforce size and injury outcomes across countries and economic sectors. The results show a strong positive association between workforce size and total accident counts. However, this relationship becomes substantially weaker when normalized indicators are used, confirming that employment-adjusted rates provide a more accurate representation of underlying occupational risk.

Taken together, these exploratory results highlight several structural characteristics of occupational injury patterns in the Transportation and Storage sector. These patterns provide the analytical foundation for the interpretation presented in the following section, where their implications for operational risk management are discussed.

Figure 4.4 - Relationship Between Workforce Size and Workplace Injuries (EU Top-10, 2014–2023)



A strong positive relationship exists between workforce size and total accident counts, indicating that absolute injury volumes are largely driven by exposure.

This relationship weakens substantially when injury incidence is normalized by employment, confirming that employment-adjusted indicators are necessary to assess underlying occupational risk.

5. Key Findings and Interpretation

This section synthesizes the main insights emerging from the exploratory data analysis of work-related injuries in the Transportation and Storage sector (NACE Rev. 2 – Activity H). The objective is to **interpret the patterns** identified in the previous section and highlight their **implications for understanding occupational risk within the sector**.

Key Takeaways

- **Injury incidence in the Transportation and Storage sector has remained relatively stable over the 2014–2023 period,** with no clear long-term trend in normalized accident rates.
- **Cross-country differences in absolute accident counts largely reflect variations in workforce size,** while normalized indicators show broadly comparable risk levels across major European economies.
- **Workplace injuries tend to concentrate in recurring operational contexts,** particularly transport operations, goods handling, and vehicle-related tasks.
- **The sector workforce is structurally male-dominated,** contributing to the higher share of male casualties observed in injury statistics.
- **Within the Italian economy, the Transportation and Storage sector records higher injury incidence than the national average,** confirming its classification as a relatively high-risk activity.

5.1 Stability of Injury Incidence Over Time

Has occupational injury risk in the sector changed over time?

Injury incidence in the Transportation and Storage sector remained broadly stable over the 2014–2023 period.

While moderate year-to-year fluctuations are observed, the normalized indicator of casualties per 1,000 employees does not display a persistent long-term trend. This suggests that the underlying occupational risk profile of the sector has remained relatively constant throughout the period analyzed.

5.2 Cross-Country Differences and Workforce Size

Why do accident counts differ across European countries?

Differences in absolute accident counts across countries are largely explained by variations in workforce size rather than substantial differences in risk intensity.

When accident figures are normalized by employment exposure, injury incidence across major European transportation and storage economies appears more comparable. In this context, Italy remains broadly aligned with the peer-country group and does not emerge as a structural outlier in relative injury incidence.

5.3 Operational Contexts of Workplace Injuries

In which operational situations do most injuries occur?

Workplace injuries tend to concentrate within a limited number of recurring operational contexts.

In particular, accidents frequently occur in situations related to transport operations, goods handling activities, and vehicle-related tasks. These patterns reflect the operational characteristics of logistics and transport activities and highlight key areas where preventive safety measures may be most effective.

5.4 Workforce Demographics and Gender Patterns

How does workforce composition influence injury patterns?

The sector workforce is strongly male-dominated, contributing to the higher share of male casualties observed in injury statistics.

This structural characteristic reflects the occupational composition of the sector and should be considered when interpreting gender differences in injury incidence.

5.5 Sector Risk Profile Within the Italian Economy

How does the sector's injury risk compare with the broader Italian economy?

The Transportation and Storage sector records higher injury incidence than the national average across economic activities.

Although the sector does not represent the most extreme risk environment within the Italian economy, it consistently remains above the overall national average and is therefore classified as a relatively high-risk activity.

5.6 Summary of Key Findings

Overall, these findings indicate that occupational injury risk in the Transportation and Storage sector is primarily shaped by **structural characteristics of logistics and transport operations**, rather than by short-term cyclical fluctuations. The concentration of accidents within specific operational contexts also highlights **clear priority areas for preventive safety interventions**.

6. Datafolio Design

6.1 Key insight

Injury risk in Transportation & Storage is **structurally high (~2.1× vs average)**.

It is not random. It is driven by a **small number of recurring operational patterns**.

This makes it:

- **Predictable**
- **Measurable**
- **Actionable**

The datafolio is built to support one goal:

- **Fast, evidence-based decisions**

It removes complexity and focuses only on what matters.

6.2 Structure (decision flow)

Context (left):
The datafolio follows a simple logic:
Context → Evidence → Actions

Context (left): What is the problem?

- Shows where risk originates
- Highlights core drivers:
  - ✓ movement
  - ✓ transport operations
  - ✓ manual handling

Output: clear definition of where to look

Actions (right): What should be done?

- Translates insights into priorities:
  - ✓ process design
  - ✓ training
  - ✓ safety routines

Output: clear direction for execution

6.3 Why these visuals

Each visual has a specific role:

- **Sector comparison** → establishes priority

- **Time trend** → shows no structural improvement
- **Accident breakdown** → isolates repeatable scenarios
- **Headline KPI** → makes scale immediately visible

No redundant charts.

Each element supports a decision.

6.4 So what

The message is simple:

- Risk is **concentrated**, not widespread
- A few scenarios drive most injuries

What to do:

- Focus on **high-frequency risk situations**
- Redesign processes where needed
- Reinforce execution through training and routines

What to avoid:

- **Broad, generic safety actions**
- Initiatives not linked to specific risk drivers.

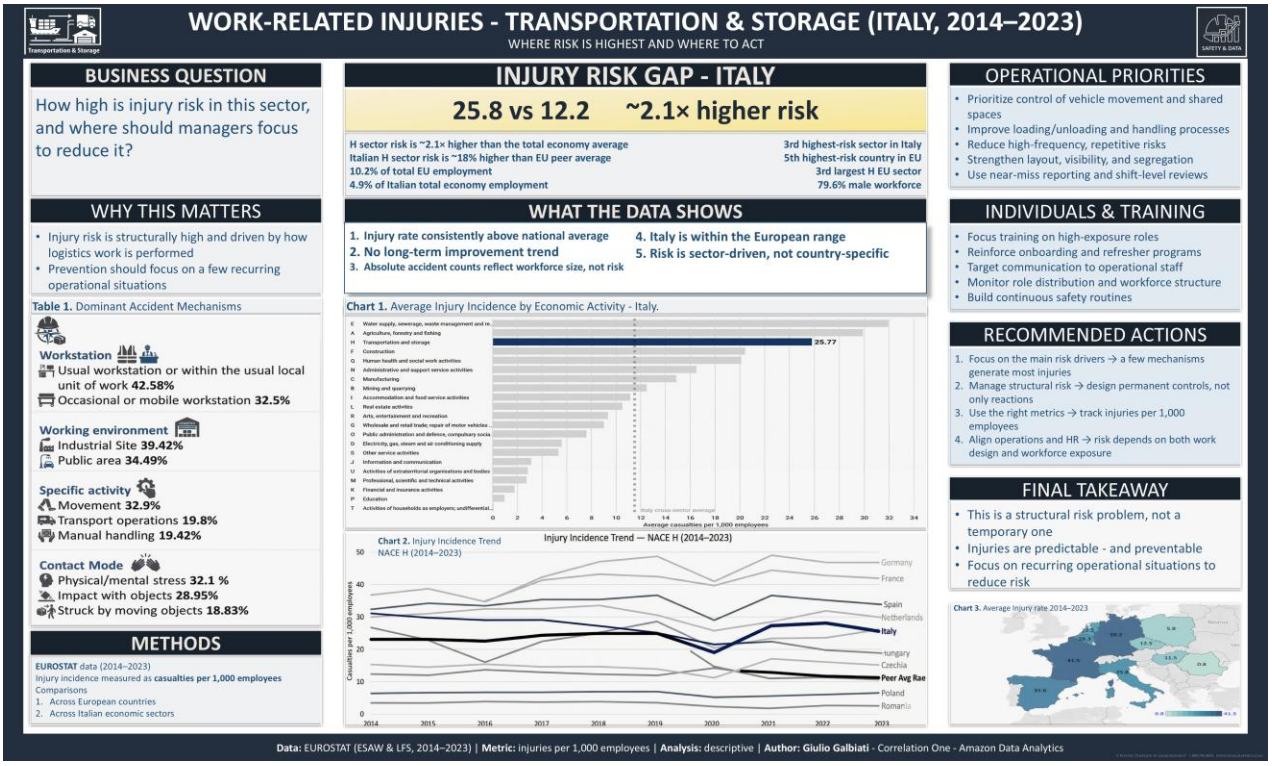
6.5 Key Takeaway

This is not a compliance problem.

It is an operational problem.

- Fix the processes where risk happens
- Focus on what repeats
- Execute consistently

Figure 6.1 - Datafolio



The Datafolio is available at the following link:

https://www.linkedin.com/feed/update/urn:li:activity:7441355004218732546/?originTrackingId=42hC6D6Qmvgs2LsbV6dWwg%3D%3D

7. Dashboard Design

7.1 Purpose

The datafolio tells the story, the dashboard shows the data behind it. Its role is simple:

- Make the analysis **transparent**
- Allow users to **validate key insights**
- Enable **drill-down into risk drivers**

This is not a storytelling tool. It is a **decision support tool**.

For management, the dashboard answers one question:

"**What is really driving risk, and where should we act?**"

7.2 Design Principles

The dashboard is built on four principles:

1. **Same metrics, no translation**

- All visuals use the same indicators as the report (injuries per 1,000 employees)
- No reinterpretation needed

2. **From overview to detail. Users can move from:**

- system view →
- country/sector view →
- operational drivers

3. **Focus on signal**

- Only structural patterns are shown
- No noise, no unnecessary breakdowns

4. **Built for decisions**

- Each chart answers a specific operational question
- If it doesn't support a decision, it's not included

7.3 Dashboard Structure

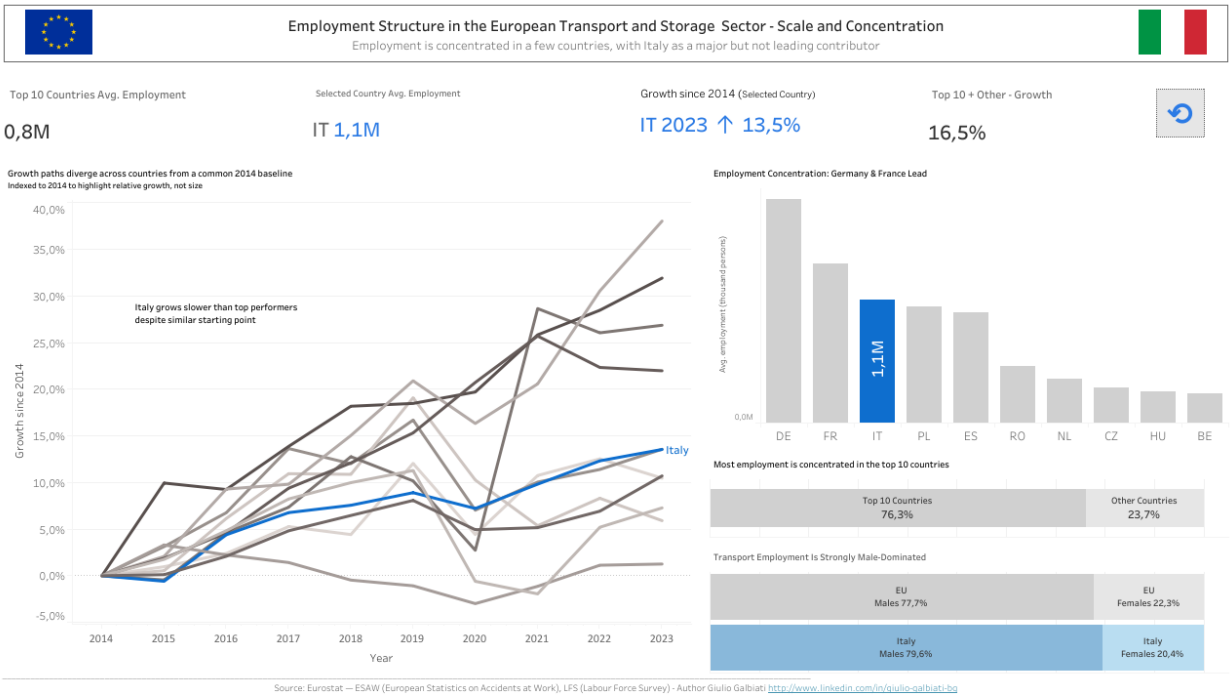
The dashboard follows a simple logic:

Context → Risk → Drivers → Focus

1. **Context: Workforce and Exposure – Dashboard 1**

   What are we dealing with?

Figure 7.1 - Employment structure (Dashboard 1)



Employment is concentrated in a few countries

- Italy is large but not leading
- Workforce is ~80% male

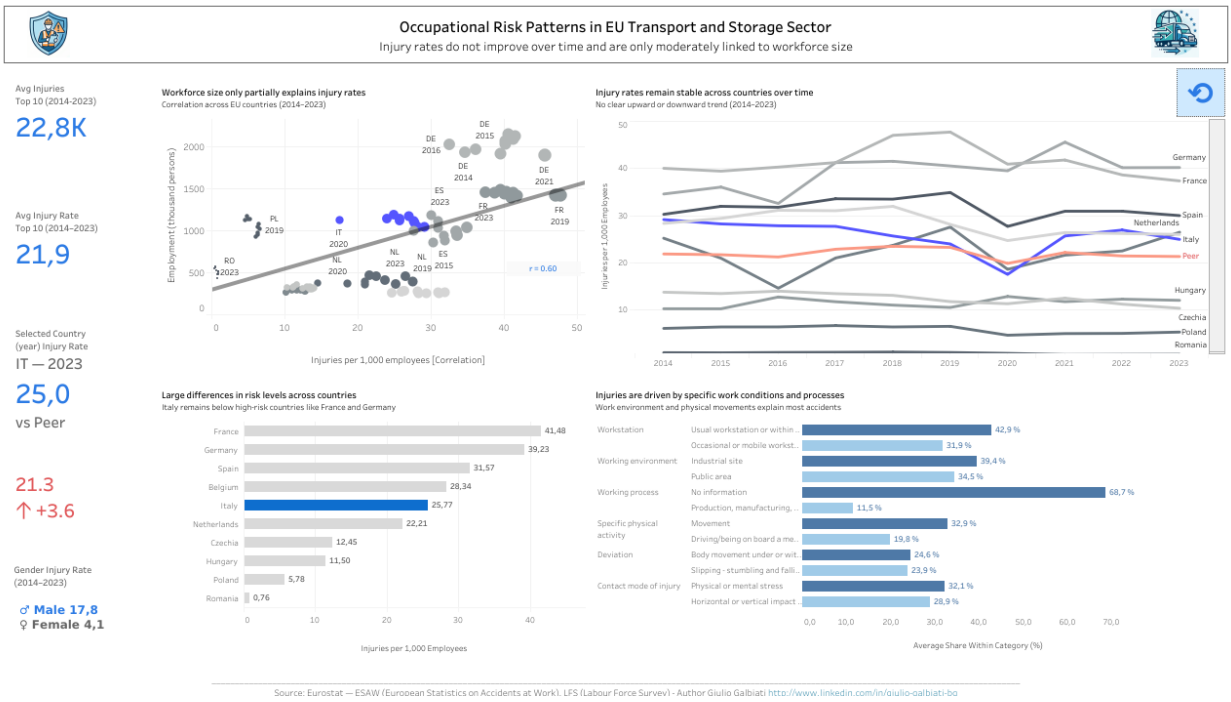
Why it matters:

- Risk must always be read against exposure.
- More people ≠ more risk.

2. Risk Patterns: What is happening (Dashboard 2)

How big is the problem?

Figure 7.2 - Occupational Risk Patterns (Dashboard 2)



- Injury incidence is **stable over time**
- Countries differ, but within a **similar range**
- Workforce size only partially explains risk ($r \approx 0.60$)

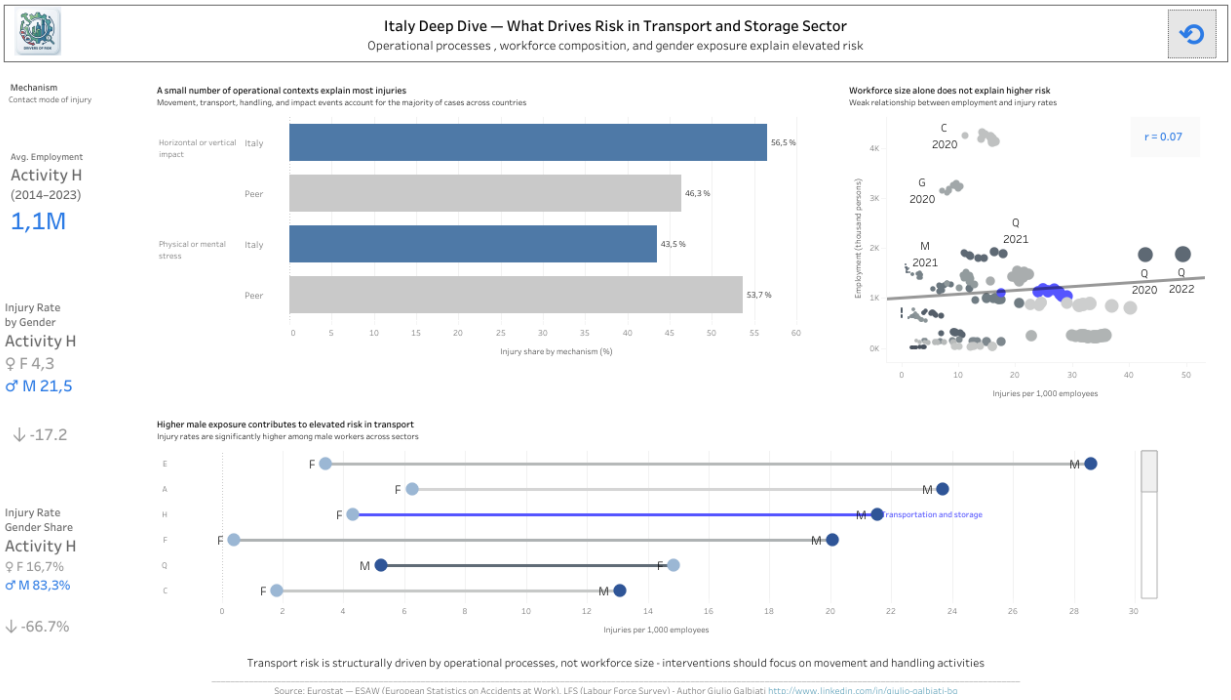
What this means:

- This is not a temporary issue
- Risk is **structural and persistent**

3. Drivers: Why it happens (Dashboard 2 and 4)

Where does risk come from?

Figure 7.3 — Italy deep dive – Risk drivers (Dashboard 4)



Most injuries come from a **small set of activities**:

- movement
- transport operations
- goods handling

Impact events and physical stress dominate

Gender differences reflect workforce structure

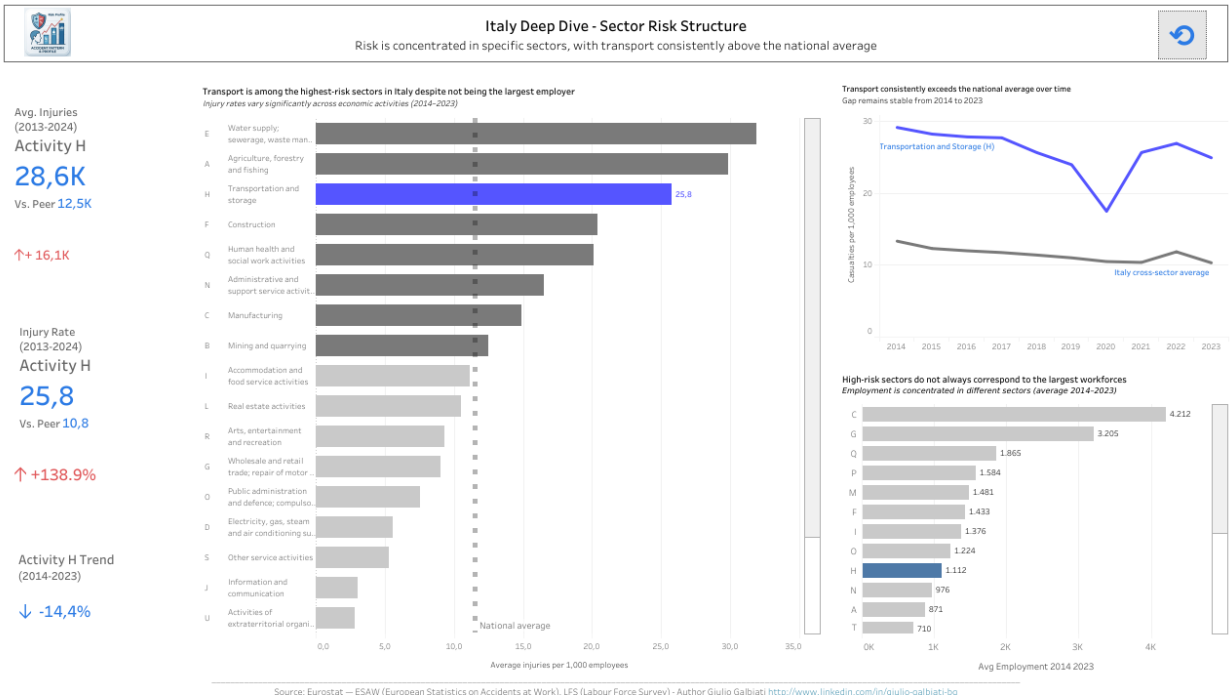
What this means:

- Risk is **concentrated and repeatable**
- This is where action should focus

4. Italy Deep Dive: Where to focus – Dashboard 3

How critical is the sector?

Figure 7.4 - Italy deep dive – Risk drivers (Dashboard 3)



Transport is **consistently above national average**

- The gap is **stable over time**
- High risk ≠ largest workforce

What this means:

- This is a **priority area**, not an outlier

7.4 Why This Design

The dashboard mirrors how decisions are made:

- **Understand exposure**
- **Size the problem**
- **Identify drivers**
- **Focus action**

Each step removes uncertainty.

7.5 Relationship with the Datafolio

The two components work together:

- **Datafolio → tells you what to do**
- **Dashboard → shows you why**

The datafolio simplifies.

The dashboard proves it.

This ensures:

- **Speed in decisions**
- **Confidence in the data**

7.6 Managerial Value

For management, this dashboard enables:

1. **Clear prioritization**

   Risk is not everywhere. It is concentrated.

2. **Targeted action**

   Focus on:

   - movement
   - handling
   - transport operations

   Not generic safety programs.

3. **Scalable insight**

   Patterns are structural and transferable to:
   - warehouses
   - transport networks
   - last-mile operations

7.7 Key Takeaway

Risk in Transportation & Storage is:

Structural → not improving over time

Concentrated → few drivers explain most injuries

Operational → driven by how work is done

This is actionable.

- Focus on a few high-frequency scenarios
- Fix processes
- Reinforce execution

Avoid broad, unfocused interventions.

The interactive dashboard is available on Tableau Public:
https://public.tableau.com/app/profile/giulio.galbiati/viz/Work-relatedinjuriesintheItaliantransportationandstoragesector/ItalyDeepDive2

8. Conclusions and Future Work

This section summarizes the main conclusions emerging from the analysis of work-related injuries in the **Transportation and Storage sector (NACE Rev. 2 – Activity H)** and outlines possible directions for further research.

Key Takeaways

- **Occupational injury risk in the Transportation and Storage sector is primarily shaped by structural characteristics of logistics and transport operations.**
- **Injury incidence remained relatively stable over the 2014–2023 period, with no persistent long-term trend in normalized accident rates.**
- **Once employment exposure is considered, injury incidence across major European sector economies appears broadly comparable, with Italy aligned with the peer-country group.**
- **Within the Italian economy, the Transportation and Storage sector consistently records higher injury incidence than the national average,** confirming its classification as a relatively high-risk activity.
- **Workplace injuries tend to concentrate in recurring operational contexts,** highlighting priority areas for preventive safety interventions.

8.1 Main Conclusions

What do the results reveal about occupational injury risk in the sector?

Occupational injury risk in the Transportation and Storage sector is primarily shaped by structural characteristics of logistics and transport operations rather than by short-term fluctuations.

The analysis shows that **injury incidence remained broadly stable over the period 2014–2023**, with moderate year-to-year variation but no persistent long-term trend.

Cross-country comparisons indicate that, **once employment exposure is considered, injury incidence among major European transportation and storage economies is broadly comparable**. In this context, Italy does not appear as an outlier and remains aligned with the European peer group.

Within the Italian economy, however, **the sector consistently records higher injury incidence than the overall national average across economic activities**. This confirms the sector's classification as a relatively high-risk environment from an occupational safety perspective.

The analysis also highlights that **accidents tend to concentrate within a limited number of operational contexts associated with transport operations, goods handling activities, and vehicle-related tasks**. These recurring patterns indicate clear areas where preventive safety interventions could be most effective.

8.2 Analytical Contribution of the Study

What value does this analytical framework provide?

The curated dataset and comparative analytical framework developed in this study provide a transparent basis for examining occupational injury patterns within the transportation and storage sector.

By combining accident statistics with employment exposure data, the analysis **enables normalized comparisons across countries and economic sectors**. This approach supports a more consistent interpretation of occupational safety conditions and provides a practical basis for evidence-based decision-making on safety priorities in logistics and transport activities.

8.3 Future Research Directions

How could the analysis be further developed?

Although the dataset includes variables related to age and injury severity, these dimensions were not fully incorporated into the core analysis due to differences in how age classes are defined in the employment and injury datasets, as well as the limited granularity of the available age categories.

These limitations prevented a **consistent comparison of injury incidence across age groups**.

Future research could extend the present analysis in several directions.

First, **resolving the incompatibility between age classifications in the available datasets and obtaining more detailed age categories** would allow a more detailed examination of how injury risk varies across different age groups. This could provide valuable insights into whether certain demographic groups are more exposed to occupational risk within the sector.

Second, **more detailed injury severity classifications** could support deeper analysis of differences between minor and severe injury outcomes.

Additional improvements could also include the integration of **occupational or task-level variables**, where available, to better isolate operational risk drivers within transport and logistics activities.

Such extensions would allow the analytical framework developed in this project to support **more detailed occupational safety assessments** and contribute to **more targeted preventive strategies** in the transportation and logistics sector.

8.4 Operational Recommendations

What actions should be prioritized based on the analysis?

The analysis indicates that occupational injury risk in the Transportation and Storage sector is **concentrated in a limited number of recurring operational contexts**. As a result, **safety interventions should be targeted** rather than broad.

Based on these findings, the following actions are recommended:

- **Prioritize high-frequency risk scenarios**, particularly those related to worker movement, goods handling, and transport operations, as these account for a significant share of injuries.
- **Conduct focused process reviews** in these operational areas to identify where risk originates within daily workflows, with particular attention to repetitive and routine tasks.
- **Redesign operational processes where needed**, especially in activities involving manual handling and interaction with vehicles or moving equipment.
- **Reinforce execution through standardized safety routines and targeted training**, ensuring that preventive measures are consistently applied in high-risk situations.
- **Avoid broad, non-targeted safety initiatives** that are not directly linked to the main drivers of injury risk identified in the analysis.
- **Monitor normalized injury incidence over time** (casualties per 1,000 employees) to evaluate the effectiveness of implemented actions and support continuous improvement.

This targeted approach maximizes the impact of safety interventions by focusing resources on the highest-risk operational contexts.

Focus on the few scenarios that drive risk, fix the processes behind them, and execute consistently.

9. Limitations

This section outlines the main limitations affecting the analysis presented in this report. These constraints arise primarily from the **characteristics of the underlying statistical sources** and from the **analytical scope** of the study.

Key Takeaways

- **The analysis relies on official statistical sources whose reporting practices may vary across countries and years.**
- **Some accident circumstance variables contain a significant share of observations classified as "Not specified" or "No information."**
- **Differences in age classifications between employment and injury datasets prevented the construction of age-standardized injury indicators.**
- **A limited number of backcasted values were introduced to maintain time-series continuity.**
- **The analytical framework is descriptive and comparative and therefore does not support causal inference or predictive modeling.**

9.1 Statistical Data Constraints

What limitations arise from the underlying statistical sources?

The curated dataset inherits several constraints from the **European Statistics on Accidents at Work (ESAW)** reporting system. Differences in national reporting practices, structural breaks in some time series, and incomplete demographic coverage may affect the consistency of observations across countries and years. These limitations were documented during the data curation phase rather than statistically imputed.

9.2 Missing or Unspecified Information

How do incomplete reporting categories affect the analysis?

Some accident circumstance variables contain a substantial share of observations classified as "**Not specified**" or "**No information**." In the Italian Transportation and Storage sector, these categories represent a notable proportion of cases in variables such as Workstation and Working Process. These values reflect limitations in reporting systems rather than meaningful operational mechanisms. As a result, interpretations based on these specific variables should be treated with caution.

9.3 Dataset Comparability Constraints

What limitations affect comparability between datasets?

Comparability constraints exist between employment and injury datasets. **Age classifications differ across sources**, preventing the construction of age-standardized injury incidence indicators. This limitation restricted the possibility of analyzing demographic exposure patterns across age groups.

9.4 Data Reconstruction and Approximation

Were any approximations introduced in the dataset?

A limited number of **backcasted values** were introduced to maintain aggregate time-series continuity. These observations are treated as approximations and are interpreted cautiously within the analysis.

9.5 Scope of the Analytical Framework

What conclusions can and cannot be drawn from the analysis?

The analytical framework adopted in this report is **descriptive and comparative**. The dataset supports the identification of structural patterns and relative risk indicators but is not intended for **causal inference, predictive analysis, or regulatory benchmarking**.

10. References & Acknowledgements

10.1 Data Sources

The analysis is based on official European statistics, as introduced in Section 2 (Data Sources).

The primary dataset is sourced from:

Eurostat — European Statistics on Accidents at Work (ESAW)

All data were directly accessed, extracted, and curated by the author. The dataset scope, definitions, and limitations are described in Section 2 and should be considered when interpreting the results.

10.2 Methodological Approach

The methodology applied in this report follows standard analytical practices commonly used in data analysis and business intelligence contexts.

The key approaches include:

- **Benchmarking** (peer comparison) to assess relative risk levels
- **Backcasting techniques** to estimate a limited number of missing values in time series data
- **Correlation analysis**, interpreted with awareness of its limitations (correlation does not imply causation)
- **Exploratory Data Analysis (EDA)** to identify patterns, trends, and anomalies
- **Data curation and preparation** to ensure consistency and reliability of the dataset
- **Data visualization and dashboard design** to support clear, decision-oriented communication (implemented using Tableau)

These methods reflect foundational analytical concepts covered during the training program delivered by Correlation One and standard publicly available statistical knowledge (e.g., general reference materials such as Wikipedia).

10.3 Use of Artificial Intelligence Tools

Purpose of AI Assistance

Artificial intelligence (AI) tools were used as a **supporting resource during the preparation of this report**. Their use was limited to assistance with **drafting, language editing, and the structural organization of the written narrative**, including suggestions aimed at improving clarity, terminology consistency, and formatting.

Analytical Independence

All analytical activities - including **data extraction, data preparation, SQL query development, indicator computation, visualization design, and interpretation of results** - were independently performed and validated by the author. AI tools were **not used to generate, modify, or process the datasets used in the analysis**.

Responsibility for the Analysis

The analysis presented in this report is based on the **curated dataset described in the Data Curation document** and on the **exploratory analysis documented in the EDA report**. All methodological choices and analytical conclusions reflect the author's independent work.

The author retains full responsibility for the **accuracy and interpretation of the results presented in this report**. AI assistance was used **only as a writing support tool and did not replace the author's analytical work**.